Public Health Disease Management Guidelines

Influenza, Avian (Includes H5N1 and H7N9)

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Health and Wellness Promotion Branch

Public and Rural Health Division

Alberta Health

Influenza, Avian | Alberta Health, Government of Alberta

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Revisions

Revision Date	Document Section	Description of Revision
December 2023	General	• Entire guideline updated to align with current information.
	Case Definition	• Probable and Suspect Case Definitions updated to incorporate components of the H5N1 case definition published by the Public Health Agency of Canada in July 2023.
	Public Health Management	• Animals added as possible source of exposure. • Additional information added to provide as part of management of a case. • Link added for Alberta's Body of Deceased Persons regulation. • Additional general preventative measures provided.
	Appendix - Management of Exposures to HPAI H5N1	• Algorithm updated to reflect recommendation to test symptomatic individuals who may have been exposed to H5N1.

Case Definition

Confirmed Case

A person with laboratory confirmation of avian influenza (AI) A infection (e.g., H5N1 or H7N9).

Probable Case

A person with clinical illness[(A)] **AND/OR** [(B)] who meets the exposure criteria, where:

- Laboratory diagnostic testing is positive for influenza A, negative for H1 and H3 by real-time reverse transcription polymerase chain reaction (RT-PCR) and therefore unable to be subtyped,[(C)]

OR

A person with clinical illness[(A)] **AND** who meets the exposure criteria, with one of the following:

- Laboratory diagnostic testing is positive for influenza A and subtype is not available or is indeterminate/unreliable (such as due to specimen quality, viral load, or timing),

  OR

- Rapid influenza diagnostic test (e.g., rapid antigen test or rapid molecular assay) is positive for influenza A (refer to Diagnosis section for more information) but subtype information is not available.

OR

A person for whom laboratory testing for H5 or H7 subtype was positive but not confirmed by the National Microbiology Laboratory (NML), (i.e., presumptive positive).[(D)]

Suspect Case

A person meeting exposure criteria and clinical illness[(A)] criteria and for whom laboratory test results are pending, unavailable, or not done.

(A) Clinical illness: signs and symptoms consistent with acute respiratory illness such as fever, sore throat, cough, shortness of breath, arthralgia, myalgia, and prostration, or conjunctivitis. Young children, the elderly and people who are immunocompromised may not develop a fever. For more information see Clinical Presentation.

(B) If the index of suspicion for A(H5N1) is high and the laboratory tests indicate that it does not appear to be seasonal flu, report the individual as a probable case, whether they meet either exposure or illness criteria, or both.

(C) This does not include samples that are unable to be subtyped as a result of low viral loads.

(D) All confirmatory testing is currently done at the National Microbiology Laboratory (NML).

Exposure Criteria

In the 10 days before onset of illness, a person with one of the following:

- Close (within 2 meters) unprotected (without use of respiratory and eye protection)[1,2][E] exposure to a person who is a confirmed, probable or suspect case of human infection with avian influenza A virus (e.g., in a household or healthcare facility),

OR

- Exposure to birds/animals presumed/confirmed to be infected with AI,[F]

OR

- Unprotected exposure to biological material in a laboratory setting (e.g., primary clinical specimens, virus culture isolates) known to contain avian influenza A virus. [3]

[E] Adequate personal protective equipment (PPE) includes:

- For individuals involved in culling or working with sick or infected birds/animals (e.g., wildlife rehabilitation centers), this is properly-fitted unvented or indirectly vented safety goggles, disposable gloves, boots or boot covers, a respirator (e.g., N95), disposable fluid-resistant coveralls, and disposable head cover or hair cover.
- For healthcare workers (HCW) this is disposable gloves, a respirator (e.g., N95), gowns and eye protection.

[F] Bird/animal exposures can include, but are not limited to:

- Handling, slaughtering, defeathering, butchering or culling of infected animals, or preparation of meat or eggs from infected animals for consumption; **OR**
- Direct contact with surfaces contaminated with feces or bird/animal parts (e.g., carcasses, internal organs) from infected birds/animals or in settings in which there have been mass animal die offs in the previous six weeks due to avian influenza A virus; **OR**
- Visiting a live poultry/animal market with confirmed bird/animal infections or associated with a confirmed/probable case of human infection with avian influenza A virus.

Reporting Requirements

Physicians

- Physicians shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed, probable or suspect AI cases by fastest means possible or FMP (i.e., phone call or email) and include the following:
  - name,
  - age,
  - date of birth,
  - gender,
  - personal health number,
  - date of death, and
  - other relevant clinical/epidemiological information.

Laboratories

- All laboratories shall report all positive AI subtypes (e.g., H5 or H7) laboratory results by FMP to:
  - the MOH (or designate) of the zone, and
  - Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services (AHS) and First Nations Inuit Health Branch

- The MOH (or designate) of the Zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed, probable and suspect cases (including deaths) and include the following:
  - name,
  - age,
  - date of birth
  - gender,
  - personal health number,
  - date of death (if applicable), and
  - other relevant clinical/epidemiological information.
- The MOH (or designate) of the Zone where the case currently resides shall forward the initial *Emerging respiratory pathogens and Severe Acute Respiratory Infection (SARI) case report form* of all confirmed, probable or suspect cases (including deaths) to the CMOH (or designate) within 24 hours with the final report within one week of discharge from hospital or resolution.
- For out-of-province or out-of-country cases and/or contacts, the following information should be forwarded to the CMOH (or designate) by FMP:
  - name,
  - date of birth,
  - out-of-province health care number,
  - out-of-province address and phone number,
  - attending physician (locally and out-of-province)
  - positive laboratory report (if applicable), and
  - other relevant clinical/epidemiological information.
- The Zone MOH (or designate) shall report any human AI outbreak to Alberta Health within 24 hours of notification using the *Alberta Outbreak Reporting Form (AORF)*.
  - At this time, **one confirmed/probable case is considered an outbreak** and all AI outbreaks are considered events of public health concern.

Additional Reporting Requirements

Canadian Food Inspection Agency (CFIA)

- Under the *Health of Animals Act – Reportable Diseases Regulations*, all confirmed, probable, and suspect animal cases of AI must be reported to CFIA.
- Animal health issues associated with the source of human AI disease should be reported by the MOH (or designate) to the District Veterinarian of the CFIA. Contact information is available at inspection.canada.ca/about-cfia/contact-a-cfia-office-by-telephone/eng/1313255382836/1313256130232

Agriculture and Irrigation

- The *Reportable and Notifiable Diseases Regulation*, under the *Animal Health Act*, lists specific diseases that are classified as reportable or notifiable, and the requirement and process for reporting these diseases.
  - All confirmed, probable, and suspect animal cases of AI in pigs, poultry, and cows must be reported to **Alberta Agriculture and Forestry**.
  - Animal health issues associated with the source of human AI disease should be reported by the CMOH (or designate) to the Chief Provincial Veterinarian.

International Health Regulation Reporting

- Under the International Health Regulations (2005), enhanced reporting of AI to the World Health Organization (WHO) is mandatory.
- Alberta Health notifies the Public Health Agency of Canada (PHAC) of every confirmed or probable case of human influenza caused by a novel influenza virus (e.g., H5N1 or H7N9) and PHAC notifies the WHO.

Clinical Assessment and Epidemiology

Etiology

Avian influenza (AI) viruses are type A viruses that are classified into low pathogenicity avian influenza (LPAI) A viruses or highly pathogenic avian influenza (HPAI) A viruses. The categories refer to molecular characteristics of a virus and the virus' ability to cause disease and mortality in chickens in a laboratory setting. Most AI viruses are low pathogenic. Only some AI viruses (e.g., some H5 and H7) are classified as HPAI A viruses, while most A(H5) and A(H7) viruses circulating among birds are LPAI A viruses.(4) Both HPAI and LPAI viruses can spread rapidly through poultry flocks.(4)

HPAI and LPAI designations do not correlate with the severity of illness in cases of human infection with these viruses; both LPAI and HPAI A viruses have caused mild to severe illness in infected humans.(4) They may include, but are not limited to, H5, H7, and H9 subtypes and may involve novel N subtypes as well (e.g., N6, N9). The majority of recognized human infections with AI viruses have been caused by either highly pathogenic avian influenza A(H5N1) Asian strain or low pathogenic avian influenza A(H7N9).

The main global human health concern with AI virus is that outbreaks, particularly in domestic poultry flocks, present an opportunity for ongoing genetic mutation or viral reassortment.(5) Simultaneous infection with human influenza and avian influenza viruses in an intermediary host (including mammals such as pigs as well as humans), may provide an opportunity for an exchange of genes. One possible outcome is the emergence of a novel influenza virus subtype able to infect people easily and spread from person-to-person in an efficient and sustained way that could trigger an outbreak, or pandemic.

Clinical Presentation

Human infections from influenza A virus subtypes H5, H7 and H9 range from eye infections (conjunctivitis) to influenza-like illness (ILI) symptoms to severe respiratory illness.(6) In some humans infected with A(H5N1) viruses, symptoms have rapidly progressed to severe pneumonia, acute respiratory distress syndrome (ARDS), multiple organ failure, lymphopenia, elevated liver enzyme levels and abnormal clotting profiles, diarrhea, vomiting, abdominal pain, pleuritic pain, bleeding from the nose and gums, shock and even death.(6,7)

Diagnosis

Diagnosis of all influenza subtypes are mainly made through a variety of molecular assays (e.g., polymerase chain reaction [PCR]). Routinely used molecular assays are typically designed to detect an influenza infection but not necessarily to differentiate all subtypes of influenza, and the former would need to be validated for any emerging strain of novel influenza. Avian subtypes pose a challenge as the subtyping assays that are routinely used are designed for identification of the circulating seasonal subtypes (i.e., H1 and H3) instead of the zoonotic subtypes, which then requires an additional work-up such as H5- and H7-specific molecular assays and sequencing.

Rapid molecular assays (e.g., rapid RT-PCR) detect influenza viral RNA or nucleic acids in upper respiratory tract specimens in approximately 15-30 minutes.(8) Some molecular assays are able to detect and discriminate between infections with influenza A and B viruses, others can identify specific seasonal influenza A virus subtypes, for example A(H1N1)pdm09, or A(H3N2), and others can detect influenza viruses and other respiratory pathogens (e.g., SARS-CoV-2). There are a number of rapid molecular assays (i.e., rapid RT-PCR) available in Canada (e.g., Abbott ID NOW™ Influenza A & B 2 test, Cobas® SARS-CoV-2 & Influenza A/B Test); however, they may not distinguish beyond A/B strain type.(9,10)

Rapid influenza diagnostic tests (RIDTs) are immunoassays that qualitatively (positive/negative) identify the presence of influenza A and B viral nucleoprotein antigens in respiratory specimens.(11) However, RIDTs cannot distinguish between avian and seasonal influenza and therefore have limited value in the laboratory confirmation of AI at present.(11)

Treatment

- Decisions on treatment protocols will depend on severity and should be determined by the clinician, in consultation with an infectious disease specialist, as required.
- In order to be most effective, antiviral drugs such as oseltamivir, zamanavir or peramivir should be administered as soon as a diagnosis of AI is considered; do not delay treatment while waiting for specimen collection.
- Supportive care (e.g., ventilation, dialysis) may also be required.
  - Published resources for treatment of AI: Government of Canada[12] and U.S. Centers for Disease Control and Prevention (CDC).[13]

Reservoir

Wild aquatic birds (e.g., waterfowl such as ducks, geese, swans, gulls, and terns, and shorebirds, such as storks, plovers, and sandpipers) are natural reservoirs of all influenza A subtypes.[4]

AI viruses can affect all species of birds (poultry, exotic and pet birds, and wild birds), although some species are more resistant to infection than others. AI viruses do not normally infect mammalian species; however, cases in humans and other species have been reported (e.g., raccoons, skunks, foxes, cats and dogs).[14]

Transmission

The majority of AI cases (e.g., H5N1) have been reported after direct exposure to infected birds/animals, mainly infected poultry.[12] Infected birds/animals may shed AI virus through their saliva, mucous and feces. Human infections with AI viruses can occur through contact with mucous membranes (e.g., eyes, nose or mouth) or inhalation. Mammal-to-human influenza A (H5N1) transmission is theoretically possible; however, no cases have been documented.

Some AI cases have been reported after exposure to contaminated environments such as poultry farms or live bird markets.[12] Although rare, human cases of AI (H5N1) possibly associated with consumption of raw or undercooked contaminated poultry products like raw duck organs and duck blood have also been reported.

Family clusters have been reported but almost always appear to be related to common exposures; however, limited human-to-human spread through close contact could not be excluded.[6] Human-to-human transmission is considered rare.

Incubation Period

The incubation period for AI A(H7N9) has been reported as 5 to 6 days, with a range from 1 to 15 days.[G][5] The incubation period for AI A(H5N1) has been reported as 1 to 5 days, and up to 7 days.[15]

Period of Communicability

The period of communicability for AIs is unknown. However, it is presumed that infected individuals can shed the virus from the day before symptoms begin and are considered infectious until symptoms resolve. Compared to human influenza viruses, cases with H5N1 disease have been reported to have detectable viral RNA in the respiratory tract for a longer period.[16]

[G] For the purpose of case identification and follow up of contacts, the available evidence supports exposure criteria based on 10 days prior to illness onset. This is considered a reasonable approximation with some loss of surveillance sensitivity balanced against the consideration of public health capacity to conduct public health investigation and follow-up of cases and contacts.

Host Susceptibility

Susceptibility is universal. Generally, there is little or no natural immunity to AI.(17) Young adults make up a large proportion of AI cases and the highest case fatality rate of AI is among persons age 10 to 39 years.(18) This may be due to higher risk of exposure such as occupational exposure (e.g., poultry worker, live animal market) or with live poultry kept in shared living space.

Incidence

The WHO monitors AI worldwide:

- See: Avian influenza (who.int).

The Government of Canada monitors:

- Influenza in people via FluWatch: www.canada.ca/en/public-health/services/diseases/flu-influenza/influenza-surveillance.html.
- AI in domestic poultry (reportable disease): inspection.canada.ca/animal-health/terrestrial-animals/diseases/reportable/avian-influenza/hpai-in-canada/status-of-ongoing-avian-influenza-response-by-prov/eng/1640207916497/1640207916934.
- The Canadian Wildlife Health Cooperative (CWHC) wild bird AI testing: www.cwhc-rcsf.ca/avian_influenza_testing_results.php.

The Government of Alberta monitors:

- Influenza in people, including avian and other novel influenza viruses:
  - www.alberta.ca/stats/influenza/influenza-statistics.htm (current season) and
  - open.alberta.ca/publications/2561-3154 (previous seasons).
- AI in poultry (reportable disease): www.alberta.ca/avian-influenza-reportable.aspx.
- AI in wild birds (reportable disease): https://www.alberta.ca/avian-influenza-in-wild-birds.

In 2014, a fatal case of H5N1 in an Alberta resident with recent travel to China was reported to Alberta Health as a case of novel influenza.(19) In Alberta, AI has been reportable since 2022. No other human cases of AI have been reported in Alberta.

Public Health Management

Key Investigation

- Confirm that the individual meets case definition.
- Obtain history of illness, including date of onset of signs and symptoms and determine incubation period and period of communicability.
- Determine the possible source of infection, considering the incubation period, reservoir and mode of transmission. Assess confirmed, probable or suspect cases for ALL of the following possible **exposures in the 10 days before onset of illness**:[5,15]
  - **Resided in or returned from** an area where human cases of AI have recently been detected or where AI is known to be circulating (including other areas of Canada or Alberta),
  - **Close contact** with one of the following:
    - a person who has confirmed, probable or suspect AI infection, or
    - an ill person (i.e., respiratory illness) who resided in or returned from an area where AI is known to be circulating (including other areas of Canada or Alberta).
  - **Occupational risks**, which include but are not limited to:
    - Health care worker (HCW)[H] providing direct care for a confirmed, probable or suspect AI case;
    - Laboratory setting – working with live AI virus;
    - Working directly with live, dead, or recently killed birds/animals including, but not limited to:

      Poultry farm,
      Poultry processing plant,
      Culling poultry (catching, handling, transporting, or disposing of dead birds),
      Wild bird/animal rescue/rehabilitation,
      Live bird/animal market,
      Pet industry,
      Chef working with live or recently killed domestic poultry or wild game, and/or
      Veterinary work.
  - **Bird/animal exposures** include:[5,7]
    - touching infected birds/animals,
    - touching or breathing in dust containing feces or other secretions of infected birds/animals,
    - preparing infected poultry meat for consumption if safe food handling practices are not observed,
    - slaughtering, defeathering, handling carcasses of or butchering infected birds/animals,
    - handling infected birds/animals for sale, and/or
    - attending markets that sell live birds/animals.
  - Assess use of personal protective equipment (PPE) and duration of exposure for all occupational exposures.
- Identify **close contacts**.

[H] HCWs include all health practitioners and all individuals at increased risk for exposure to, and/or transmission of, a communicable disease because they work, study, or volunteer in one or more of the following health care environments: hospital, nursing home (facility living), supportive living accommodations, or home care setting, mental health facility, community setting, office or clinic of a health practitioner, clinical laboratory.

Table 1: Close Contact Definition

CLOSE CONTACT[20,21]
Individual with: • Unprotected[I] exposure within approximately 6 feet (2 meters) of a confirmed, probable or suspect human case while the case was infectious (i.e., beginning 1 day prior to illness onset and continuing until resolution of illness). ***Note:*** *This may include but is not limited to:* - *Anyone who provided care for a confirmed, probable, or suspect case, including HCW,[E] family members or other caregivers,* - *Others who have had similar close physical contact (kissing, shared cigarettes, food, glasses/bottles, eating utensils) with a confirmed, probable, or suspect case, or* - *Persons who lived with or stayed overnight with a confirmed, probable, or suspect case.*

[I] An individual may be considered unprotected if at the time of the exposure they did not consistently and appropriately use personal protective equipment (PPE).[28]

Management of a Case

- Hospitalized cases should be isolated in accordance with hospital infection prevention and control guidelines.
- For non-hospitalized cases, consider conducting **active daily monitoring** (e.g., telephone contact) of the individual's health status for duration of illness depending on the severity of the AI strain and reasonable resource allocation.[5,20]
- For non-hospitalized cases or cases well enough to be discharged from hospital and continue isolation at home, provide the following information:[5,20]
  - Isolate away from others in the home as well as from animals (both domestic and wild) until 24 hours after symptom resolution, or, if known, for the duration of the infectious period of the specific strain,
  - Illness care in the home, including practicing respiratory etiquette and hand hygiene; limiting close and direct contact with others inside/outside the home; wearing a mask if unable to isolate from others; improving indoor ventilation; frequently cleaning and disinfecting high-touch surfaces.
    - Designate one caregiver to provide direct care to the case. Family members caring for the case should take precautions when interacting with the case and limit contact with others outside the home.
  - When/where to go for medical assessment if symptoms worsen, and instruct case to disclose recent diagnosis of AI prior to arriving at a health care setting (e.g., by telephone),
  - How to prevent transmission, and
  - Importance of and how to access seasonal influenza immunization.

Body of Deceased Persons

- All remains of deceased human cases of AI must be handled in accordance with Alberta's Bodies of Deceased Persons Regulation. For additional information, refer to the Routine Practices and Additional Precautions for Preventing the Transmission of Infection in Healthcare Settings.

Management of Contacts (of Human/Bird/Animal AI Cases, or Contaminated Environments)

See Appendix: Management of Exposures to HPAI H5N1 Infected Bird/Animals and/or Contaminated Sites

In general, the public health management of contacts of a human case, infected bird(s)/animal(s) or a contaminated environment and the decision to offer post-exposure prophylaxis is based on an assessment of the exposure and should be reviewed by the Zone MOH in consultation with the CMOH and others (e.g., CFIA, Alberta Agriculture and Irrigation) as needed.

Table 2: Exposure Risk[5,21,22]

Level of Risk	Type of Exposure
High-risk exposure groups (Recognized risk of transmission)	• Individual(s) with unprotected[J] close exposure[K][21] to a flock[23] or large group[L] of sick or dead birds/animals infected with AI or to particular birds/animals that have been directly implicated in confirmed/probable human AI cases. • Individual(s) with unprotected[J] close exposure[K] to sick birds/animals or while decontaminating affected environments (including bird disposal) as part of AI outbreak control efforts (e.g., personnel involved in culling sick/infected birds/animals). • Individuals involved in the handling and slaughtering of live poultry and other animals, such as in a live animal market, in an affected area or visitors to an area where such activities are being undertaken while unprotected.
Moderate-risk exposure groups (Unknown risk of transmission)	• Close contact of a confirmed, probable, or suspect human AI case. • Individual(s) with unprotected[J] close exposure[K] to single or small groups of sick or dead birds/animals infected with AI in an outdoor location which is not densely populated by animals of the same species as the infected animal (e.g., single wild bird in a park). • HCW(i.e., those working in a setting where health care is being provided) who had no, or insufficient PPE in place when in close contact (i.e., within two metres) of a confirmed, probable, or suspect human AI case, or in direct contact with respiratory secretions or other potentially infectious specimens from the case. • HCW or laboratory personnel who had unprotected contact (i.e., did not have or was wearing insufficient PPE) with specimens/secretions which may contain virus or with laboratory isolates.
Low-risk exposure groups (Transmission unlikely)	• HCW who used appropriate PPE during contact with human AI cases (i.e., in the absence of significant human-to-human transmission). • HCW working in a unit/office where a confirmed, probable, or suspect human AI case(s) is diagnosed but is **not** a close contact or had no direct or indirect contact with infectious material from that case(s). • Others who have had social contact of a short duration with a confirmed, probable or suspect case in a non-hospital setting (e.g., in a community or workplace environment).[22] • Individual(s) who wore adequate protection while in close exposure[K] to sick birds/animals or decontaminating affected environments (including bird disposal) as part of AI outbreak control efforts (e.g., personnel involved in culling sick/infected birds/animals). • Individual(s) without adequate protection (e.g., PPE) who were involved in culling non-infected or likely non-infected bird populations as a control measure (e.g., those exclusively culling asymptomatic animals in a control area outside of the infected and restricted zones). • Individuals who handle (i.e., have direct contact with) asymptomatic birds that may be infected with AI based on species and possibly proximity to a geographic area where AI has recently been identified (e.g., bird banders). • Laboratory personnel working with the influenza virus using appropriate laboratory procedures and infection control precautions.

[J] Unprotected includes did not have, or wearing insufficient PPE, or had a breach in the use of PPE, during these activities.

[K] Close exposure may include contact with birds/animals (e.g., handling, slaughtering, defeathering, butchering, preparation for consumption); direct contact with surfaces contaminated with feces or bird parts (carcasses, internal organs, etc.); or prolonged exposure to birds/animals in a confined space.

[L] Flock or large group: a number of birds of one kind that are kept or fed together or are herded together. Social birds such as gulls and starlings are also often seen in large groups.

Table 3: Summary of Public Health Management for Asymptomatic Contacts Based on Exposure Risk[5]

Human illness risk	**Exposure risk** (Table 2)		
	Low risk groups (Transmission unlikely)	**Moderate risk groups** (Unknown risk of transmission)	**High risk groups** (Recognized risk of transmission)
Subtype/strain is **not known to have caused human illness**	**Public Health Measures** - Self-monitor - Education	**Public Health Measures** - Self-monitor - Education	**Public Health Measures** - Consider quarantine and offer testing - Self-monitor - Education
	PEP* – None	**PEP** – None	**PEP –** Consider prophylaxis
Subtype/strain is known to cause **predominantly mild human illness**	**Public Health Measures** - Self-monitor - Education	**Public Health Measures** - Self-monitor - Education	**Public Health Measures** - Quarantine and offer testing - Self-monitor - Education
	PEP – None	**PEP –** Consider prophylaxis	**PEP** – Offer prophylaxis
Subtype/strain is known to cause **predominantly severe human illness**	**Public Health Measures** - Self-monitor - Education	**Public Health Measures** - Daily active monitoring - Education	**Public Health Measures** - Quarantine and offer testing - Daily active monitoring - Education
	PEP – none	**PEP** – Offer prophylaxis	**PEP** – Offer prophylaxis

**PEP = Post-exposure prophylaxis*

- **Self-monitor:** for symptoms for the duration of the surveillance period.
  - The surveillance period depends on the incubation period of the AI virus and commences on the last day of exposure to the case (day 0).
- **Quarantine:** all high-risk contacts of sub-types known to cause human illness (e.g., H5N1, H7N9).
  - The quarantine period is the maximum incubation period for that sub-type and commences on the last day of exposure to the case (day 0).
- **Testing**: Offer testing for influenza and/or other respiratory pathogens, depending on current testing algorithms, *regardless of symptoms*. Timing and frequency of specimens (e.g., NP swabs) to be determined by the zone MOH in consultation with the ProvLab Virologist-On-Call and CMOH.
- **Education – Provide the following information**:[20]
  - **Monitor** for the appearance of fever and other influenza-like-illness symptoms, which include acute onset of respiratory symptoms, sore throat, arthralgia, myalgia, and prostration. Young children, the elderly and people who are immunocompromised may not develop a fever.
  - **Isolate immediately** if symptoms develop within the surveillance period and **notify** public health and health care provider.
    - Symptomatic persons should seek medical attention even if they are only mildly ill. Notify care provider/facility of exposure history in advance of presenting for care and ensure appropriate use of PPE.
  - **Defer travel** for the duration of surveillance period for those not under quarantine. This entails remaining within a reasonable distance of home, family physician and/or local hospital to facilitate rapid assessment and isolation if symptoms develop and to facilitate active surveillance by public health.
  - **Limit contact** with cases or symptomatic persons in the home.

- **Perform hand hygiene** such as washing hands with soap and water frequently or using alcohol-based hand sanitizers (containing at least 60% alcohol) when soap and water are not available or when hands are not visibly dirty.
- **Ensure regular cleaning and disinfection** of high-touch objects and surfaces.
- **Offer seasonal influenza vaccine** immediately (as seasonally available) to those who have not received the most recent annual influenza vaccine.

Post-Exposure Prophylaxis (PEP)

Chemoprophylaxis is the only PEP currently available for AI. PEP is provided to prevent severe disease and to limit secondary transmission.

- For persons in high-risk exposure groups, PEP may be considered or offered depending on the avian influenza subtype.
- For persons in moderate-risk exposure groups, decisions to initiate antiviral chemoprophylaxis should be based on risk assessment and clinical judgment, with consideration given to the type of exposure risk (see Table 2: Exposure Risk), time since exposure, duration of exposure and to whether the close contact is at higher risk for complications from influenza.[M][21,22,24]
- For persons in low-risk exposure groups, PEP is not recommended.
- If PEP is initiated, treatment frequency dosing for the oseltamivir or zanamivir (twice daily) is recommended instead of the typical seasonal influenza antiviral PEP regimen (once daily).[5,23]
  - Administration of PEP should begin as soon as possible (within 48 hours) after the first exposure to the confirmed or probable case or infected birds/flocks. Delayed PEP increases the risk of severe disease and secondary transmission.
  - Duration of PEP: 5 or 10 days depending on exposure. If the exposure was time-limited and not ongoing, the recommended duration is 5 days from the last known exposure. If the exposure is likely to be ongoing (e.g., household setting), a duration of 10 days is recommended.
  - More information on antiviral recommendations for avian influenza is available at: https://www.canada.ca/en/public-health/services/publications/diseases-conditions/guidance-human-health-issues-avian-influenza.html#a11
- If the decision is to offer PEP to close contacts, Public Health should direct them to their primary health care provider for the antiviral prescription.

[M] People at higher risk for complications/hospitalizations from influenza include:

- people who are pregnant,
- children and adults with the following chronic health conditions:
  - cardiac or pulmonary disorders
  - diabetes mellitus and other metabolic diseases
  - cancer and other immune compromising conditions due to underlying disease and/or therapy
  - renal disease
  - anemia or hemoglobinopathy
  - neurological or neurodevelopmental conditions
  - morbid obesity (body mass index of 40 and over)
  - children up to 18 years of age undergoing treatment for long periods with acetylsalicylic acid (ASA)
- residents of nursing homes and other chronic care facilities,
- adults 65 years of age and older,
- all children younger than 60 months of age,
- people who have reduced access to health care, such as vulnerable populations like those experiencing homelessness or those with disabilities,
- people who are at an increased risk of illness because of living conditions, such as those:
  - living in shelters who experience overcrowding,
  - who have limited access to facilities for personal hygiene,
  - persons with disabilities who might have limited capacity to understand or perform personal hygiene practices.

Outbreak Management

One confirmed/probable case of human AI is considered an outbreak. AI outbreaks are managed on a situation specific basis as per direction from the zone MOH, in consultation with the CMOH and others (e.g., CFIA, Alberta Agriculture and Irrigation) as needed.

Travel

In the absence of efficient human-to-human transmission, and with no evidence of transmission occurring during travel, tracing contacts who were in proximity of cases who were symptomatic during a flight or while on other conveyances (e.g. train, bus) is not recommended at this time.[20]

Preventive Measures

- While there is currently no AI vaccine licensed for use in Canada, there are a number of vaccine candidate viruses that could be used to produce a vaccine if needed.[25,26]
- While seasonal influenza immunization will not prevent AI infection, it may prevent co-infection. Preventing co-infection is intended to prevent possible viral recombination in an individual infected with both seasonal and AI strains, which could theoretically result in a novel pandemic strain.[5]
  - All Albertans six months of age and older are eligible to receive annual seasonal influenza vaccine. Refer to the Alberta Immunization Policy for more information.
- Pneumococcal vaccine may be useful in preventing secondary bacterial infections in populations at high risk for influenza-related complications.
- General and ongoing education for the public:[5]
  - Avoid handling live or dead wild birds or other wildlife.
  - If an AI virus known to be of risk to human health is confirmed in a specific area, individuals should avoid exposure to known or potential sources of AI virus (e.g., wild birds, other susceptible wildlife, feces, or AI contaminated environments).
  - Individuals on farms where AI has been confirmed in birds/animals, who are not directly involved in culling or cleaning/disinfection activities, should avoid exposure to sick or dead animals, litter, manure, or contaminated surfaces where possible.[5]
  - If travelling to an affected area, follow the precautions below:[7]
    - Avoid high-risk areas (e.g., poultry farms, live animal markets, areas where poultry may be slaughtered and contact with any surfaces that appear to be contaminated with feces from poultry or other animals);
    - Avoid high-risk activities (e.g., handling sick or dead birds); and
    - Practice food safety and good hygiene practices (e.g., hand washing with soap and water).
  - Monitor health during/after travel:
    - See a health care provider if flu-like symptoms develop while travelling or upon returning to Canada, and
    - Make health care provider aware of travel or living in an area where AI (e.g., H5N1) is a concern.
  - More information on avian influenza and human health can be found here.

Appendix - Management of Exposures to HPAI H5N1 Infected Birds/Animals and/or Contaminated Sites

Background

- The HPAI H5N1 viruses circulating globally in 2021 and 2022 appear to be different than previously detected HPAI H5N1 viruses. The current circulating strains do not infect people easily or cause severe illness in humans. Based on information available at this time, it is thought that current H5N1 viruses represent less of a risk to human health than earlier H5N1 viruses.[27]
- Since April 2022, HPAI subtype H5N1 virus has been detected in commercial poultry and wild birds in Alberta.
- The same strains have also been found in many other parts of the world, including in several other Canadian provinces and the United States. It has been spread primarily due to the migration of infected waterfowl.
- H5N1 has also been found in multiple mammal species in Canada and the United States, including foxes, skunks, mink, otters, bobcats, opossums, coyotes, bears, dolphins, seals, and domestic mammals such as cats and dogs.[5]
- The risk of human infection is considered to be very unlikely for the general public, and unlikely for those with occupational exposures.
- More information is available here.

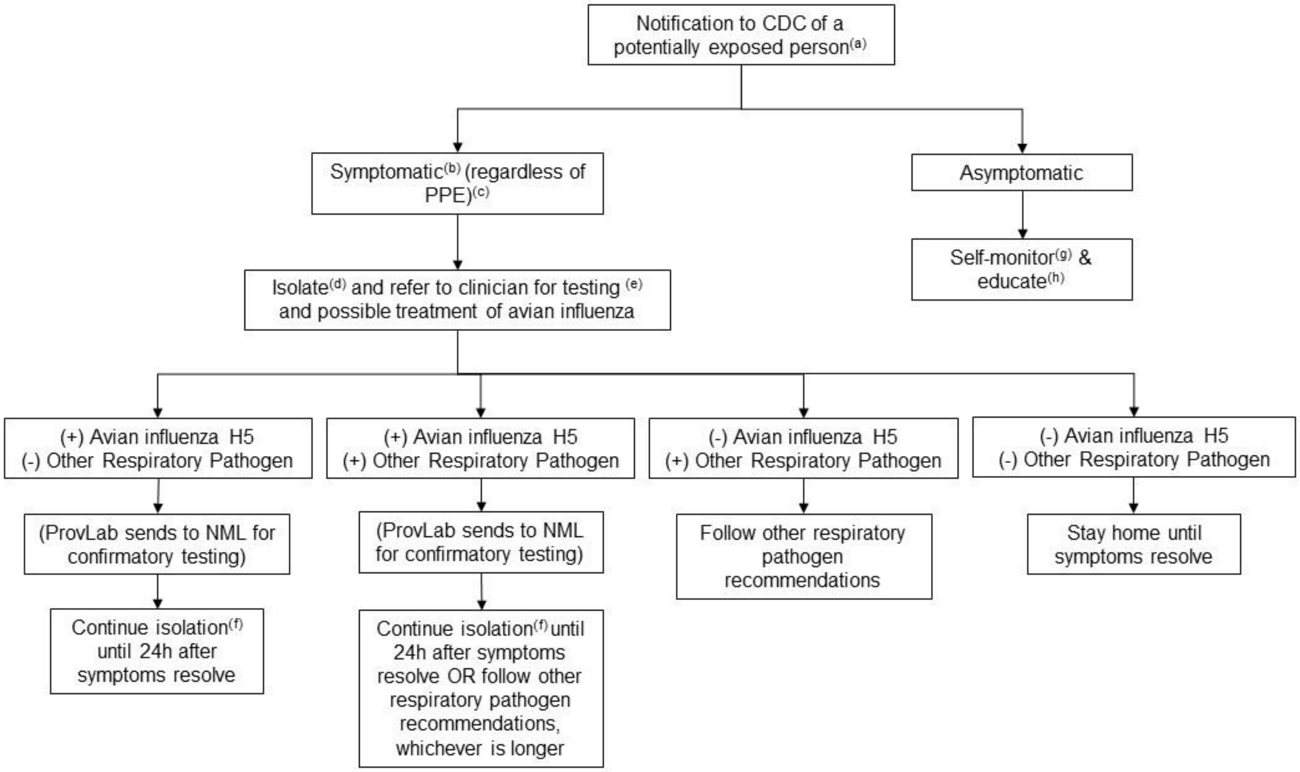
Recommended Process

- In Alberta, symptomatic individuals that had exposure to potentially infected birds/animals in the past 10 days can call Health Link at 811 for more information.
- Health Link will refer individuals to AHS CDC to assess their exposure and screen for symptoms.

Recommended Actions

- AHS CDC will assess the person for influenza symptoms and exposure risk (Figure 1: Algorithm for Assessing Symptoms and Recommendations for Testing and Treatment) and make recommendations for testing/treatment as appropriate (see Table 2: Exposure Risk):

Figure 1: Algorithm for Assessing Symptoms and Recommendations for Testing and Treatment



(a) **Exposed Person**: A person exposed to infected birds/animals and/or contaminated sites linked to known AI outbreak areas.

(b) **Symptomatic**: Fever, cough, sore throat, runny or stuffy nose, muscle and/or body aches, headaches, fatigue, conjunctivitis (red eyes), shortness of breath, and less frequently, diarrhea, nausea, or vomiting.

- Notify CMOH (or designate) by FMP (see Reporting Requirements) of any symptomatic people who have known AI exposure in the 10 days prior to symptom onset.

(c) **PPE**: It is important to assess PPE to determine risk of exposure. While adequate PPE may be deemed low risk exposure, there may be circumstances where the individual is unaware of a breach, or it is later determined that the recommended PPE was insufficient.

- **Adequate PPE (low risk exposure)**:
  - For individuals involved in bird/animal culling or working with sick or infected birds/animals (e.g., wildlife rehabilitation centers), this is properly-fitted unvented or indirectly vented safety goggles, disposable gloves, boots or boot covers, a respirator (e.g., N95), and disposable fluid-resistant coveralls, and disposable head cover or hair cover.[2]
  - For members of the public or hunters (i.e., non-occupational exposures), wear gloves or use a doubled plastic bag if needing to handle dead, sick, or injured birds and avoid contact with blood, body fluids and feces. For more information see here.
- **No or Inadequate PPE (medium risk exposure)**: absence of any of the above OR there is a concern about a possible breach.

(d) **Isolate:** Symptomatic individuals should be told to isolate immediately, if not already.

(e) **Test**: The person should be referred to their health care provider to be evaluated for testing and antiviral treatment.

- Avian influenza, SARS-COV-2, and Respiratory Pathogen Panel (RPP) testing are recommended.
- If treatment is recommended by their health care provider, it should not be deferred while waiting for their test results.
- MOH must approve specimen collection and notify the ProvLab virologist on call (VOC) that a specimen(s) is being sent.
- If household contacts develop symptoms before test results are available, they should also isolate and be advised to notify AHS CDC (as directed by AHS CDC).

(f) **Isolation for AI:** AI cases should isolate for seven days or until 24h after symptoms resolve, whichever is longer.

(g) **Self- Monitor**: Asymptomatic individuals should monitor for symptoms of AI (see b). If any symptoms occur in the 10 days following an exposure, the person should isolate immediately and call Health Link at 811.

(h) **Educate**: Provide information that current H5N1 bird flu viruses in North America are different from previous H5N1 bird flu viruses (see Background). Provide counselling that although the general risk is deemed to be low, it is still important to monitor for any symptoms for the 10 days following the last exposure as there is still a small possibility of human infection that needs to be stopped from spreading further.

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